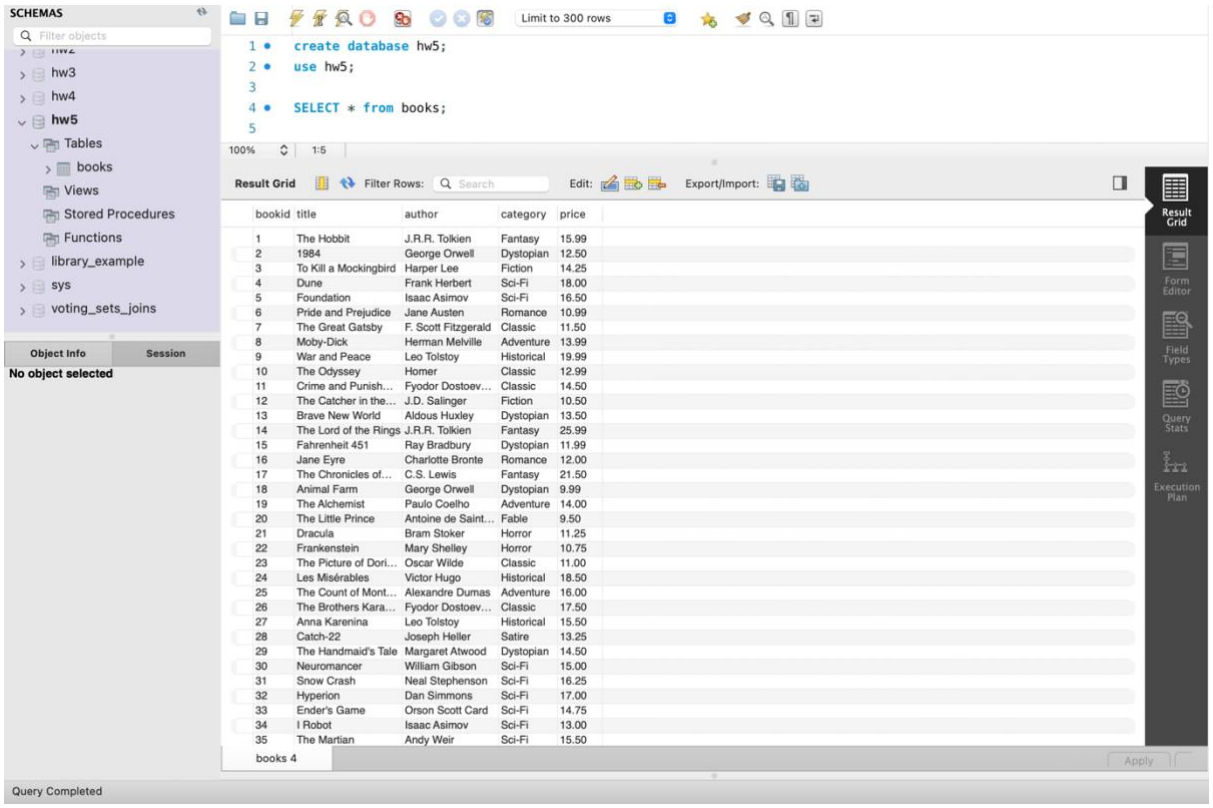
Dr. Forouraghi

CSC 351 D01

Dan Shapovalov

Homework #5





Node.js code for my server

```
//*****************************************************************************
//*** set up an HTTP server off port 3000
//*****************************************************************************
const express = require('express');
const app = express();
const PORT = 3000;
```

```
// Requires the exported pool object from the local 'mysql.js' file
const pool = require('./mysql');

// Keep book values from being interpreted as HTML tags (just additional security feature)
function escapeHtml(value) {
    return String(value ?? '').replace(/[&<>"']/g, char => ({
        '&': '&amp;', '<': '&lt;', '>': '&gt;', '"': '&quot;', "'": '&#39;'
    })[char]);
}

//*****************************************************************************
//*** receive GET requests from the browser
//*****************************************************************************
app.get('/', async (req, res) => {
    const category = req.query.category;
    const author = req.query.author;
    const format = req.query.format;

    let sql_query = 'SELECT bookid, title, author, category, price FROM books';
    let params = [];

    // Use a prepared statement for whichever search parameter was provided
    if (category) {
        sql_query += ' WHERE category = ?';
        params = [category];
    } else if (author) {
        sql_query += ' WHERE author = ?';
        params = [author];
    }

    try {
        const [result, fields] = await pool.execute(sql_query, params);

        if (format === 'text') {
            // start creating the HTML body for the browser
            let html_body = '<HTML><STYLE>body{font-family:arial}</STYLE><BODY><TABLE BORDER=1>';

            // print column headings
            html_body += '<TR>';
            for (let i = 0; i < fields.length; i++) {
```

```
            html_body += `<TH>${escapeHtml(fields[i].name.toUpperCase())}</TH>`;
        }
        html_body += '</TR>';

        // print rows of table data
        for (let i = 0; i < result.length; i++) {
            const book = result[i];
            html_body += `<TR>
                <TD>${escapeHtml(book.bookid)}</TD>
                <TD>${escapeHtml(book.title)}</TD>
                <TD>${escapeHtml(book.author)}</TD>
                <TD>${escapeHtml(book.category)}</TD>
                <TD>${escapeHtml(book.price)}</TD>
            </TR>`;
        }

        html_body += '</TABLE></BODY></HTML>';
        return res.send(html_body);
    }

    // JSON is the default response, including when format=json.
    res.json(result);
  } catch (err) {
    console.error('SQL Error:', err);
    res.status(500).json({ error: 'An error occurred while fetching books.' });
  }
});

// server waits for incoming requests
app.listen(PORT, function () {
  console.log('NodeJS app listening on port ' + PORT);
});
```

Examples

http://localhost:3000/?author=Isaac%20Asimov&format=json



http://localhost:3000/?category=Sci-Fi

